

Progress Report

- Increment 3 -

Group #16

1) Team Members

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2) Project Title and Description

The Market Signal & Research Platform (MSRP) is a full-stack financial analytics web application designed to help users analyze stock market data and evaluate trading strategies. The system allows users to retrieve historical stock data, compute technical indicators such as Simple Moving Average (SMA) and Relative Strength Index (RSI), and run long-only backtests on rule-based strategies.

3) Accomplishments and overall project status during this increment

During the third and final increment, the team focused on refining and stabilizing the platform. Rather than introducing new features, efforts were directed toward resolving outstanding bugs, improving code quality, and enhancing the overall user experience.

Key accomplishments this increment include:

- Resolved an issue where trades remained open indefinitely by adding an optional exit reason to the Trade object (strategy, stop_loss, take_profit)
- Implemented strscdxr5ategy parameter optimization
- Added a buy-and-hold benchmark to every backtest response for comparative analysis
- Corrected a miscalculation in the annualized return (CAGR) formula
- Fixed the Sharpe Ratio calculation and added Sortino and Calmar ratio metrics
- Refactored the frontend codebase into organized, modular feature files
- Implemented Bollinger Band Breakout Strategy and Mean Reversion Signal
- Added an automated API smoke test workflow via GitHub Actions
- Added troubleshooting documentation for macOS pip SSL certificate issues
- Implemented a backend startup preflight check for required dependencies
- Resolved runtime failures caused by merge conflicts in GitHub workflows
- Hardened frontend authentication by tokenizing login sessions,
- Fixed a missing root API route that previously returned a 404 error for users
- Resolved remaining Pydantic v2 deprecation warnings
- Wired MACD and RSI indicators into a fully functional trading strategy

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

The primary technical challenge encountered this increment was a localhost connectivity issue stemming from a commit that inadvertently overwrote the frontend connection port. This was resolved by reverting the commit, restoring functionality.

In terms of scope, the team made a deliberate decision to forgo the planned AI/ML feature additions. We focused on stability, correctness, and a polished user experience over introducing new capabilities.

5) Team Member Contribution for this increment

Katya Serechenko

- Progress Report
- Source Code – Fixed frontend-backend contract issues in App.jsx and stock related backend pieces, as well as added backtests.
- Issues tackled in Github: #49, #46, #44, #56, #55, #60
- Video – 1/3 participants

Logan Bramblett

- RD document
- Source Code – Refactoring frontend files into modular files. Added workflows to github actions to prevent merge conflicts.
- Video – 1/3 participants
- Issues tackled in Github: #52, #50, #48, #47, #46, #45, #56

Dascia Pylypenko

- Source Code - Strategies, optimizers, equations, and metric enhancements. Implemented unit tests.
- IT document
- Video – 1/3 participants
- Issues tackled in Github: #58, #57, #54, #53, #51, #43, #42
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Narandan Miller

- Source Code – Added edge cases, fixed details in backend logic and added missing front-end pieces. Added ingest validation.
- Issues tackled in Github - #66-85

6) Stakeholder Communication

Dear Stakeholders,

We are pleased to share an update on the Market Signal & Research Platform (MSRP.)

In this final stage, we have successfully been able to polish the platform and prepare it for launch. We focused on detail-oriented work in regard to ensuring the logic is tested and implemented perfectly as well as providing the user a refined and clean experience.

In addition to refinement, we have introduced safeguards to our development workflow, including automated testing pipelines and merge conflict protections, to reduce the risk of critical errors and ensure the long-term stability of the platform.

After careful consideration, we have decided to defer the development of additional AI/ML-powered features to a future phase. As we approach launch, we determined that stability and reliability take precedence over

feature expansion. We remain enthusiastic about exploring these capabilities once the platform is fully established.

We look forward to discussing the next steps for MSRP and thank you for your continued support and confidence in our work.

Best regards,
The MSRP Team

- 7) **Link to video**
<https://youtu.be/i6i56L5aUa0>